

Lithium Niobate Polarization-Independent Modulator Using Integrated Polarization Splitters and Mode Converters

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Abstract—Compact lithium niobate polarization-independent electrooptic modulators using integration of polarization splitters and mode converters are presented. Experimental results show polarization-dependent loss as low as 0.04 dB can be achieved for devices fabricated on Z-cut substrates with an applied voltage of 26 V. The electrode lengths required for electrooptic modulation are 10 and 6 mm for TE and TM modes, respectively, and the total device length is only 1.6 cm. Moreover, only one diffusion step is required that greatly simplifies the fabrication process.

Index Terms—Electro-optic modulation, lithium niobate, mode converter, polarization dependent loss, polarization independent modulator, polarization splitter.

I. INTRODUCTION

LITHIUM niobate (LiNbO_3) electro-optic (EO) modulators have been very useful for high-speed communication systems [1], [2]. However, lithium niobate modulators are polarization dependent due to the efficient use of a specific electro-optic coefficient. It is then important to have polarization independent modulators in LiNbO_3 . Various device configurations have been presented for this purpose [3]–[17]. The simplest way is to use two sets of electrode configurations to modulate the horizontal and the vertical field components independently [3]. However, electrode alignment is a critical factor in the construction of the polarization-independent modulator. Moreover, as both modes are supported in the same waveguide, fringing fields in each set of electrode configuration affect the modulation, which causes polarization dependent losses. Though stepped $\Delta\beta$ reversal electrode can be used to improve the device performance, the electrode configuration becomes too complicated to be useful for high frequency application [4], [5].

Polarization independent modulation can also be realized by using X-cut Z-propagation lithium niobate substrate [6]–[9].

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However, the electro-optic coefficient used for modulation is much smaller, which causes the electrode length too long to be efficient for high frequency application. Alternatively, a special design of electrode location for polarization independent modulation was proposed [10]. Tolerance of electrode and waveguide alignment is too critical for the device to be realized. Polarization-independent intensity modulation may also be fabricated by using a lithium niobate phase modulator in a fiber Sagnac interferometer configuration [11]. The reported polarization dependent loss was found to be less than 0.25 dB at low frequency operation and less than 2 dB at frequencies higher than 20 GHz. Recently, highly stable polarization independent Mach-Zehnder interferometer utilizing displaced Sagnac configuration has been reported [12]. However, these measurement setups are not integrated devices.

An alternate way for polarization independent modulation in lithium niobate was realized by splitting polarized waves into TE and TM modes by a set of two rutiles and then converting the split TE mode into TM mode by a wave plate. Then, both TM modes are combined for electro-optic modulation [13]. Though polarization dependent loss as low as 0.3 dB was reported, many discrete parts are required. Later, the polarization-insensitive operation of lithium niobate Mach-Zehnder interferometer with a silica waveguide polarization beam splitter was reported [14]. Similarly, the guided TE modes were rotated by a wave plate such that all optical signals are aligned with the TM polarization. In their work, a polarization-dependent loss of 0.1 dB has been reported; however, hybrid integration technology must be used in order to benefit from the design flexibility offered by silica planar light circuits. For reference, a review of lithium niobate Mach-Zehnder interferometer switch applications in optical fiber networks can be found in [15].

Recently, in order to have larger vertical and lateral index contrast for better mode confinement such that TE and TM modal field distribution can be more similar, lithium niobate waveguides are fabricated by using ion milling on insulator chips [16]. However, the fabricated waveguides are highly lossy due to the sidewall roughness caused by redeposition of lithium niobate during etching process. Some more work is then required to remove the roughness. Moreover, the use of advanced fabrication techniques such as electron beam lithography and ion milling are not cost effective. A more direct way is to fabricate single-mode polarization-independent waveguides in lithium niobate

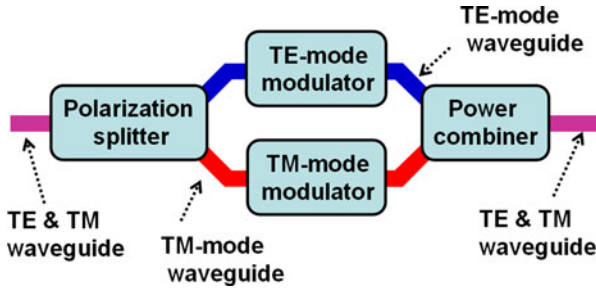


Fig. 1. Schematic diagram of the proposed polarization independent modulator.

crystal by femtosecond laser pulse irradiation [17]. That's because irradiation of femtosecond laser pulses in lithium niobate can be used to form depressed cladding waveguides [18] such that better optical confinement can be achieved. However, no details of the application for electro-optic modulation were reported.

In this work, the concept of splitting TE and TM modes [13], [14] is realized by using a polarization splitter [19]. The split TE and TM modes are then separately modulated using TE and TM mode converters [20]. The proposed polarization independent modulator does not use any discrete parts such as half-wave plate, rutile, the prism, etc., and therefore, is very compact, reliable, and less prone to damage from vibration and shock. To the best of our knowledge, the present device structure is firstly demonstrated. Experimental results show polarization dependent loss as low as 0.04 dB can be achieved when the applied voltage is 26 V.

II. DESIGN CONSIDERATION

The basic concept of the proposed polarization independent modulator is illustrated as shown in Fig. 1. The device structure is essentially an integration of a TE-TM polarization (mode) splitter, two modulators, and a TE-TM modal power combiner. The waveguide structure looks like that of a conventional Mach Zehnder interferometer except the split waveguide branches support only TE and only TM polarization as indicated in the figure. The laser is launched into the input waveguide that supports both TE and TM modes. Then, these two modes are split and directed to TE-mode and TM-mode intensity modulators. Finally, both modes are recombined at the output waveguide. When the polarization of input laser is varied, the split TE and TM modal power ratio will also be changed and modulated accordingly. As the total output power (sum of the power of both modes) is not affected, the performance of the proposed device can be independent of the polarization of the input laser.

In this work, the polarization splitter used is the same as that reported [19]. The modal power combiner is essentially a reversed polarization splitter, and therefore can be easily fabricated at the same time. Consider the TE-mode intensity modulator, which is realized by using a mode converter with a waveguide supporting only TE modes (TE waveguide). The converted modal power (i.e., TM modal power) can not be guided and is finally lost in the substrate when a modulation

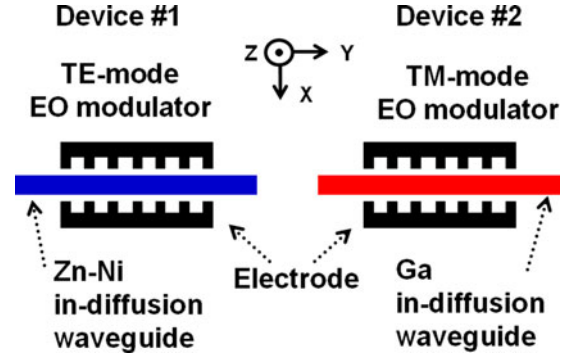


Fig. 2. Schematic diagram of the proposed single-polarization electro-optic intensity modulator.

voltage is applied. Thus, the mode converter with a TE waveguide becomes an electro-optic intensity modulator for the TE mode. Similarly, that with a waveguide supporting only TM modes (TM waveguide) can be used as a TM-mode electro-optic intensity modulator. Note that the intensity modulation principle is based on mode conversion not mode interference like that of a conventional Mach Zehnder interferometer. Note also, the electro-optic coefficient of LiNbO_3 used for mode conversion is r_{51} rather than the largest electro-optic coefficient r_{33} . Though r_{33} is not used, r_{51} is about 91% of r_{33} , which is much better than those based on using r_{22} (about 22% of r_{33}). Thus, the device can be shorter than those reported previously [8], [9].

To demonstrate the feasibility of using TE-TM mode converter with single polarization waveguide for electro-optic intensity modulation, straight gallium in-diffusion (GI) [21] and zinc-and-nickel in-diffusion (ZNI) waveguides (Fig. 2) that supports only TM modes and TE modes, respectively, are fabricated on a Z-cut lithium niobate substrate [19]. For single mode operation at 1550 nm, the waveguide width W is chosen to be $8 \mu\text{m}$. Because of the large birefringence of LiNbO_3 , mode conversion efficiency is highly wavelength dependent, which requires a pair of finger-type electrodes of period satisfying the phase matching condition,

$$\frac{2\pi}{\lambda} \times |n_{\text{TE}} - n_{\text{TM}}| = \frac{2\pi}{\Lambda}, \quad (1)$$

where λ is the input laser wavelength, Λ is the period of finger-type electrode, and n_{TE} and n_{TM} are the refractive indexes of TE and TM modes, respectively. For LiNbO_3 , the estimated difference of modal indexes at 1550 nm is 0.0734 [22]. The calculated period of finger-type electrode Λ is $21 \mu\text{m}$ [23], [24]. For easy alignment, the electrode gap d is chosen as $12 \mu\text{m}$, so the $8\text{-}\mu\text{m}$ -waveguide can be placed between the electrodes with a margin of $2 \mu\text{m}$ on both sides. To use r_{51} for electro-optic modulation, the electrode structure of the converter is shown in Fig. 3.

To demonstrate the feasibility of the proposed modulators shown in Fig. 2, preliminary experimental work is required. For simplicity, the details of the fabrication process are described later. The measurement setup is illustrated as shown in Fig. 4. A 1550 nm distributed-feedback laser is coupled to the test sample

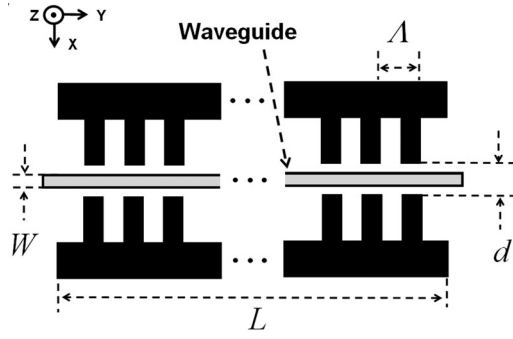


Fig. 3. Electrode structures of LiNbO₃ single-polarization electro-optic intensity modulators.

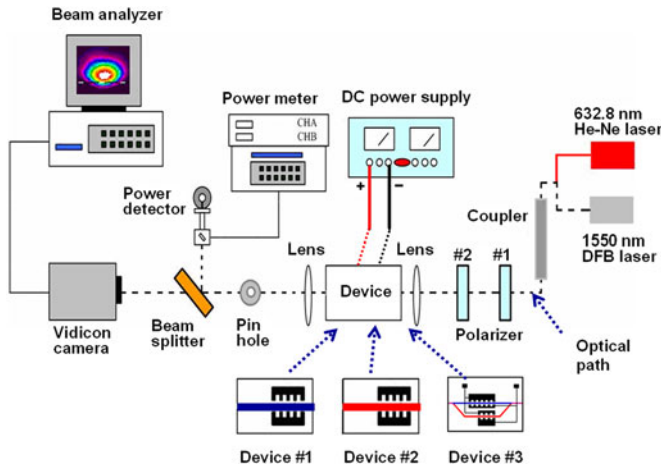


Fig. 4. Measurement setup for Device #1, TE-mode intensity modulator, Device #2, TM-mode intensity modulator, and Device #3, polarization independent modulator.

with the aid of a 632.8 nm He-Ne laser for accurate alignment. The first polarizer is used to equalize the input power of the TE and TM modes, and the second polarizer is used to select a correct polarization (TE or TM) for the input laser. The lenses are used for end fire coupling. The output optical power is detected by a power meter and the modal field contours are captured by a vidicon camera and displayed on a monitor. To characterize the proposed single-polarization electro-optic intensity modulator, the modulation efficiencies η_{TE} and η_{TM} are defined as given by

$$\eta_{TE}(V) = \left(1 - \frac{P_{TE}(V)}{P_{TE}(0)}\right) \times 100\%, \quad (2)$$

$$\eta_{TM}(V) = \left(1 - \frac{P_{TM}(V)}{P_{TM}(0)}\right) \times 100\%, \quad (3)$$

where $P_{TE}(V)$ and $P_{TM}(V)$ are output powers of TE and TM waveguides as functions of the applied voltage, respectively. Obviously, larger percentages imply more TE/TM powers being converted to TM/TE powers, and therefore have higher modulation efficiencies. The measured modulation efficiencies of the proposed modulators with a pair of electrodes of length L as a function of applied voltage V are shown Fig. 5. As can

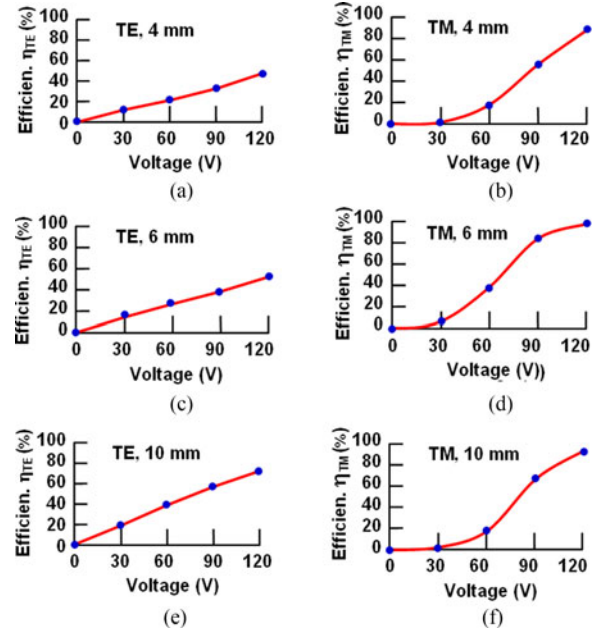


Fig. 5. Measured modulation efficiencies of the proposed TE and TM electro-optic intensity modulators (Device #1 and #2) with various electrode lengths. (a) TE, 4 mm, (b) TE, 6 mm, (c) TE, 10 mm, (d) TM, 4 mm, (e) TM, 6 mm, and (f) TM, 10 mm.

be seen from Figs. 5(a)-5(c), the modulation efficiencies of TE waveguides are increased with the applied voltage. And the longer the electrode length is, the better the efficiency is. For the TM waveguides, the results are quite similar except the best modulation efficiency is found when the electrode length is 6 mm rather than a longer (10 mm) or shorter (4 mm) one. That's because the intensity modulation is based on mode conversion, where the higher efficiency occurs when the phase matching condition is better satisfied. Moreover, the modulation efficiency is quite linear for the TE waveguides. To have high modulation efficiency, an electrode length of 10 mm is chosen for the TE-mode waveguide. For the TM modes, the linear regions of modulation efficiency are slightly smaller. To be compatible with that of the TE waveguide, an electrode length of 6 mm is chosen for the modulator.

The details of the proposed TE-TM polarization independent modulator are illustrated in Fig. 6, which is essentially an asymmetric Mach Zehnder interferometer consisting of input and output branches supporting both TE and TM modes and two branches supporting single TE-polarization and single TM-polarization modes. To ensure the mode stability, short waveguide segments of the same length 0.5 mm are added before and after the electrodes. The total device length is then about 1.6 cm. As the index change of the TM branch, fabricated by gallium in-diffusion, is greater than that of the TE branch, fabricated by zinc-and-nickel in-diffusion, the TM branch is chosen to be bent from the input waveguide at an angle of 0.5° to reduce bending losses [19]. The modal power combiner is just like a reversed polarization splitter, and therefore, all the design considerations are the same as before.

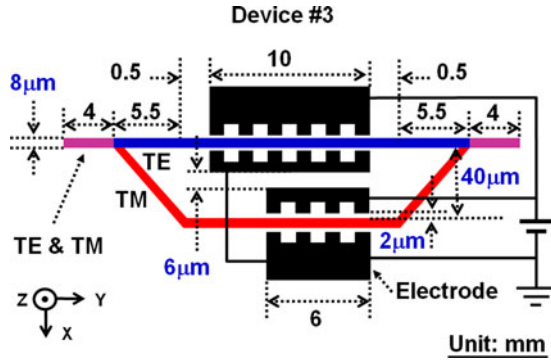


Fig. 6. Device structure of the proposed polarization independent modulator on Z-cut LiNbO_3 . All units are in millimetres unless otherwise indicated.

TABLE I
METAL FILM THICKNESSES FOR THE PROPOSED POLARIZATION INDEPENDENT ELECTRO-OPTIC MODULATOR

Component	Mode	Film Thickness ($\text{\AA}$)			
		Ni	Zn	Ga_2O_3	Al
Straight Waveguide	TE	50 + 150	750		
Bent Waveguide	TM			1000	
Input Waveguide	TE	50 + 350	750 + 1500		
Electrode	TM				3000

III. EXPERIMENT

The waveguide fabrication process parameters for different components of the proposed polarization splitter are listed in Table I. The overall fabrication process is illustrated in Fig. 7. First, a clean LiNbO_3 chip of suitable size ($1 \times 1.6 \text{ cm}^2$) is chosen (Fig. 7(a)) and a photoresist layer is spun on for lift-off (Fig. 7(b)). The region for GI waveguide is exposed and developed as shown in Fig. 7(c). Later, a Ga_2O_3 layer of thickness 1000 $\text{\AA}$ is deposited on the substrate by radio-frequency sputtering (Fig. 7(d)), which is then defined by the second mask to reveal the bent waveguide branch by lift-off technique as shown in Fig. 7(e). Another layer of photoresist is coated as shown in Fig. 7(f). With the previously-defined Ga_2O_3 waveguide pattern as an alignment mark for the third mask, the straight waveguide is defined as shown in Fig. 7(g). Then a Ni, Zn, and Ni composite layer of thicknesses 50 $\text{\AA}$, 750 $\text{\AA}$, and 150 $\text{\AA}$, respectively (first Ni-Zn-Ni layer) is sequentially thermally deposited on the substrate (Fig. 7(h)). The first Ni layer of thickness 50 $\text{\AA}$ is used to improve the adhesion of Zn layer to LiNbO_3 surface. To fabricate the input waveguide for both TE and TM modes, the central region is covered by a piece of glass of width 22 mm (Fig. 7(i)). Alternatively, the central region can also be covered by two pieces of glass of widths less than the precise width, and independently aligned to the input and output waveguides. Though the central region of the glasses has to be overlapped, the final result is the same as that using only one piece of glass during

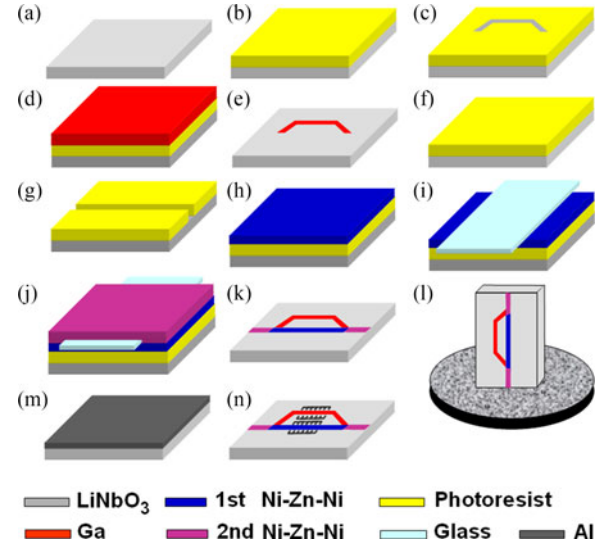


Fig. 7. Fabrication process steps of the proposed polarization-independent electro-optic modulator. (a) Choose a clean LiNbO_3 chip, (b) spin on a photoresist layer, (c) expose to ultraviolet light and develop the exposed region, (d) deposit a Ga layer, (e) lift-off the photoresist, (f) spin on a photoresist layer, (g) expose to ultraviolet light and develop the exposed region, (h) deposit Ni, Zn, and Ni layers sequentially, (i) cover the branching region with a piece of glass, (j) deposit Ni, Zn, and Ni layers sequentially, (k) remove the glass, lift-off the photoresist, and diffuse all deposited metals, (l) polish waveguide edges, (m) deposit an Al layer, (n) etch and develop the electrode pattern.

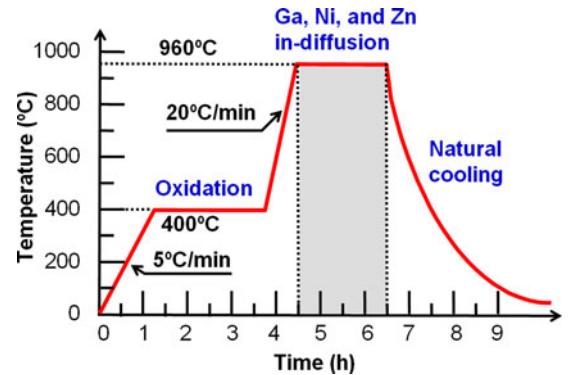


Fig. 8. Diffusion temperature vs. diffusion time used for the proposed device.

thermal evaporation. Consequently, this way is more flexible for an accurate alignment. Then, a second Ni, Zn, and Ni composite layer of thicknesses 50 $\text{\AA}$, 1500 $\text{\AA}$, and 350 $\text{\AA}$, respectively (second Ni-Zn-Ni) is deposited like that for the previous step (Fig. 7(j)). Just as the previous step, the composite layer is lifted off to reveal the waveguide pattern as shown Fig. 7(k). Note that increase of the Zn layer thickness may increase the index change for a better optical confinement of the fabricated waveguide; however, that will also increase the difficulty of lift-off. Some trade-offs are required in choosing a suitable Zn thickness. The Zn, Ni, and Ga one-step diffusion is then started in the furnace with a temperature rising curve, including oxidation, diffusion, and natural cooling, as illustrated in Fig. 8. As the melting point of Zn is 419.5 $^{\circ}\text{C}$, the heating at 400 $^{\circ}\text{C}$ for 2.5 h as indicated in Fig. 8 is for the oxidation of Zn, which is necessary to prevent

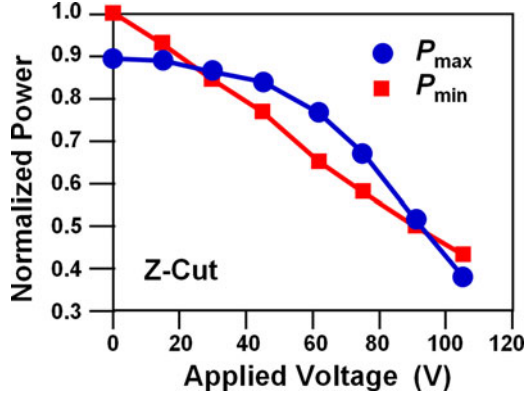


Fig. 9. Measured normalized power of Device #3 as a function of applied voltage.

TABLE II
MEASURED OUTPUT POWER AND CALCULATED POLARIZATION DEPENDENT LOSS FOR THREE APPLIED VOLTAGES

Applied Voltage (V)	$P_{\max}$ (μ W)	$P_{\min}$ (μ W)	α_{PDL} (dB)
26	537	532	0.041
63	475	406	0.680
95	289	286	0.045

the Zn strip from being distorted owing to abrupt temperature rise. After the diffusion step, both end faces of the substrate are polished for end-fire coupling. Finally, an aluminum (Al) layer of thickness 300 nm is deposited and etched to develop the electrode patterns. Note that as the adhesion of Al layer to LiNbO₃ is poor, the lift-off technique used for Ga, Ni, and Zn previously is not suitable for Al in this case.

IV. RESULTS

The measurement setup is also illustrated as shown in Fig. 4. However, in this case, the second polarizer is adjusted to measure the maximum and minimum output powers $P_{\max}$ and $P_{\min}$, respectively. The polarization dependent loss α_{PDL} is then defined as

$$\alpha_{\text{PDL}} = 10 \log \left(\frac{P_{\max}}{P_{\min}} \right), \quad (4)$$

The contours of output fields are coupled into a camera and analyzed by a beam analyzer. The measured normalized power as a function of applied voltage is shown in Fig. 9. As can be seen, both $P_{\max}$ and $P_{\min}$ are decreasing with the applied voltage. That's because the converted modal power is dissipated in the substrate and no longer contribute to the total output power. The corresponding polarization dependent loss is calculated and shown in Fig. 10. Obviously, the minimum polarization dependent losses are found when the applied voltages are 26 and 95 V, respectively. However, the maximum polarization dependent loss is occurred when the applied voltage is 63 V. The measured output power and the calculated polarization dependent loss for these three applied voltages are listed in Table II. For comparison, the measured output field intensity contours for

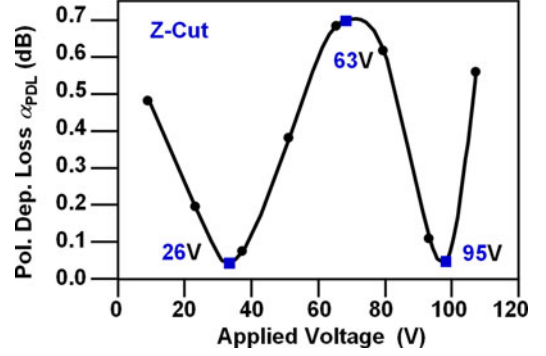


Fig. 10. Measured polarization dependent loss of Device #3 as a function of applied voltage.

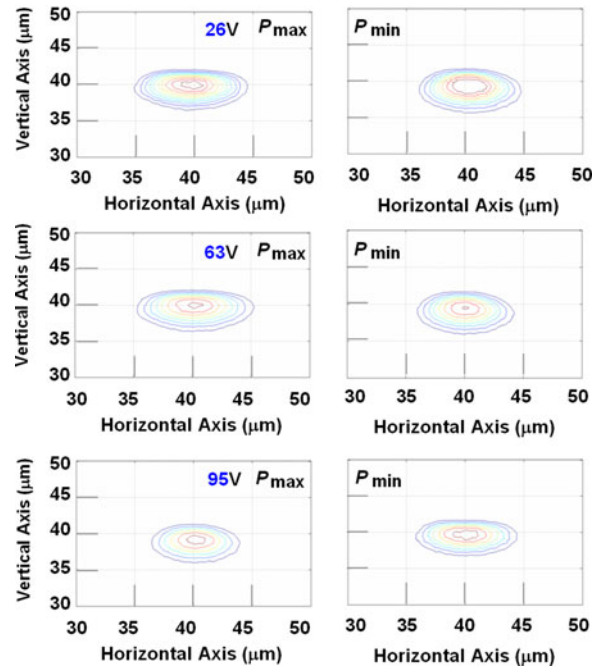


Fig. 11. Measured output field intensity contours for three applied voltages.

these three applied voltages are shown in Fig. 11. For the cases of minimum α_{PDL} (applied voltage = 26 and 95 V), the field intensity contours of TE and TM modes are almost the same, so the polarization dependent losses can be very low as expected. For the case of maximum α_{PDL} (applied voltage = 63 V), the field intensity contours of these modes are slightly different, but not much. Therefore, the corresponding α_{PDL} is a little higher than those of the previous cases.

V. DISCUSSION AND SUGGESTIONS

The extinction ratios of polarization splitters are 27.5 dB for the TE-mode, and 24.3 dB for the TM-mode [19]. These high extinction ratios cause the splitting of TE and TM modes nearly complete and can be modulated independently. Therefore, the measured polarization dependent losses can be very low as

required. Thus, to design and fabricate a good polarization splitter is essential for successful operation of the proposed device.

To reduce the applied voltage for electro-optic intensity modulation, ridge structure can be used because the applied field can be more confined in the interaction region [23]–[25]. The simplest way for making a ridge is using proton-exchange wet etching [26]. However, the ridge height may not be high enough for practical use. To improve the wet etching, field-assisted technique [27], joint proton source [28], and gamma-ray irradiation [29] were reported. Obviously, dry etching such as reactive ion etching and inductively-coupled plasma etching can also be used to fabricate the ridge structure, however, the etch rate achieved is not suitable for low drive voltage modulators [30]. Moreover, to enhance the electro-optic interaction, extended sidewall electrodes may also be deposited to reduce the drive voltage [31].

As mode converters can also be fabricated on X-cut [32] or Y-cut [33] LiNbO₃ substrates, it is then of interest to fabricate the proposed polarization independent modulator on these substrates. For example, on the X-cut substrate, in order to take the advantage of r_{51} , a pair of interdigital finger-type electrodes must be used. From theoretical study, both the electrode width and spacing must be reduced when the period is chosen to be the same. The effort spent for the fabrication of a precise electrode pattern is then increased. Thus, the improvement and application of the proposed polarization independent modulator will be a challenging and interesting work in the future.

VI. CONCLUSION

Polarization-independent electro-optic modulators using integration of polarization splitters and mode converters on LiNbO₃ substrates are successfully fabricated. For applied voltages of 26 V and 47 V, the polarization dependent losses of the proposed device can be as low as 0.04 dB. The electrode lengths required for electro-optic modulation are 10 and 6 mm for TE and TM modes, respectively, and the total device length is only 1.6 cm, so the proposed device is very compact. Moreover, as only one diffusion step is required, the device fabrication process is greatly simplified. Finally, improvements of the performance by using ridge structures are suggested for future studies.

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